Spring 2019

Marmara University
Electrical and Electronics Engineering
STAT2056-Probability and Random Variables
Final Exam

1. The input X to a communication channel is either "-1"or "1" with respective probabilities 1/4 and 3/4. The output of the channel Y is equal to: the corresponding input X with probability 0.95; $-X$ with probability 0.03; 0 with probability 0.02

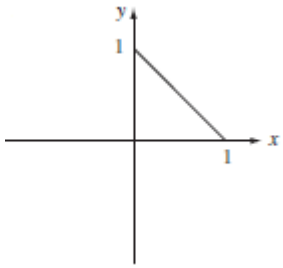
   (a) Find the joint probability mass function (PMF) $p_{X,Y}[i,j]$. [15 pts.]
   (b) Compute $P[X \neq Y]$ [5 pts.]
   (c) Find the covariance of X and for Y [5 pts.]

2. The voltage V across a 5-ohm resitor is a Gaussian random variable with mean 1 and variance 2.

   (a) Find the probability density function (PDF) of the power dissipated (Hint: $P = V^2/5$). [15 pts.]
   (b) Calculate the probabiliy that the power dissipated is less than 3 Watts. [10 pts.]

3. The continuous random vector (X, Y) has the PDF $p_{X,Y}(x,y) = k$ in the region shown below

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   (a) Find the value of k. [10 pts.]
   (b) Find the marginal PDF for X and for Y. [10 pts.]
   (c) Calculate $P[X > 0, Y > 0]$. [5 pts.]

4. The number of defects X in a VLSI chip is a Poisson random variable with the corresponding PMF

$$P[X = k] = \frac{R^k}{k!}e^{-R}, \qquad k = 0,1,\cdots$$

   However, R is itself a gamma random variable with the corresponding PDF

$$p_R(r) = \frac{\lambda^\alpha}{\Gamma(\alpha)}r^{\alpha-1}e^{-\lambda r}$$

   Find expected number of defects $E[X]$ (Hint: $E[X] = E\big[E[X|R]\big]$). [25 pts.]

Directions:

- One A4-sized formula sheet (without solutions) can be used.
- The exam is 75 minutes.

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1. a) X is a Binomial random variable with $p=0.05$, $M=100$. Y is also a Binomial random variable with $p=0.05$, $M=100$. Since X and Y are independent, we have

(15) $P_{X,Y}[X=i, Y=j] = P_X[X=i] \cdot P_Y[Y=j] = \binom{100}{i}(0.05)^i (0.95)^{100-i} \cdot \binom{100}{j}(0.05)^j (0.95)^{100-j}$, $0 \le i \le 100$, $0 \le j \le 100$

b) $P[X>5, Y>5] = P[X>5] \cdot P[Y>5] = (1 - P[X \le 5]) \cdot (1 - P[Y \le 5])$

(5) $= \left[1 - \sum_{i=0}^{5} \binom{100}{i}(0.05)^i (0.95)^{100-i}\right] \cdot \left[1 - \sum_{j=0}^{5} \binom{100}{j}(0.05)^j (0.95)^{100-j}\right]$

c) $P_X[X=i] = \sum_{j=0}^{100} P_{X,Y}[X=i, Y=j] = \binom{100}{i}(0.05)^i (0.95)^{100-i} \underbrace{\sum_{j=0}^{100} \binom{100}{j}(0.05)^j (0.95)^{100-j}}_{\sum_j P_Y[Y=j] = 1}$ (5)

$= \binom{100}{i}(0.05)^i (0.95)^{100-i}$, $0 \le i \le 100$

Because of symmetry, $P_Y[Y=j] = \binom{100}{j}(0.05)^j (0.95)^{100-j}$, $0 \le j \le 100$.

2. (a) $P = \frac{V^2}{5}$ has two solutions $v_0 = \sqrt{5p}$ and $v_1 = -\sqrt{5p}$, and $\frac{dp}{dv} = \frac{2v}{5}$. (5)

$$P_P[p] = \sum_{k=0}^{1} P_V(v) \cdot \frac{1}{\left|\frac{dp}{dv}\right|}\Big|_{v=v_k} = P_V(\sqrt{5p}) \cdot \frac{1}{\left|2\frac{\sqrt{5p}}{5}\right|} + P_V(-\sqrt{5p}) \cdot \frac{1}{\left|2 \cdot \frac{-\sqrt{5p}}{5}\right|}$$ (5)

$$= \frac{1}{2}\sqrt{\frac{5}{p}}\left[P_V(\sqrt{5p}) + P_V(-\sqrt{5p})\right]$$

$$= \frac{1}{2}\sqrt{\frac{5}{p}}\left[\frac{1}{\sqrt{8\pi}} e^{-\frac{1}{8}(\sqrt{5p}-1)^2} + \frac{1}{\sqrt{8\pi}} e^{-\frac{1}{8}(\sqrt{5p}+1)^2}\right]$$ (5)

$$= \frac{1}{2}\sqrt{\frac{5}{8p\pi}}\left[e^{-\frac{1}{8}(\sqrt{5p}-1)^2} + e^{-\frac{1}{8}(\sqrt{5p}+1)^2}\right], \quad p \ge 0.$$

(b) $P[P<3] = \int_0^3 P_P[p]\,dp$ (the student is not required to evaluate the integral because it is not a trivial task!) (10)

3. (a) $\int_0^1 \int_0^{1-x} P_{X,Y}(x,y)\,dy\,dx = \int_0^1 \int_0^{1-x} k\,dy\,dx = k \cdot \int_0^1 (1-x)\,dx = k \cdot \left(x - \frac{x^2}{2}\Big|_0^1\right) = k \cdot \frac{1}{2} = 1 \Rightarrow \boxed{k=2}$ (10)

(b) $P_X(x) = \int_0^{1-x} P_{X,Y}(x,y)\,dy = \int_0^{1-x} 2\,dy = 2\left(y\Big|_0^{1-x}\right) = 2(1-x)$, $0 \le x \le 1$ (5)

$P_Y(y) = \int_0^{1-y} P_{X,Y}(x,y)\,dx = \int_0^{1-y} 2\,dx = 2\left(x\Big|_0^{1-y}\right) = 2(1-y)$, $0 \le y \le 1$ (5)

(c) $P[X>0, Y>0] = \int_0^1 \int_0^{1-x} P_{X,Y}(x,y)\,dy\,dx = \int_0^1 \int_0^{1-x} 2\,dy\,dx = 2\int_0^1 (1-x)\,dx = 2\cdot\left(x - \frac{x^2}{2}\Big|_0^1\right) = 1.$ (5)

4. $E[X] = E[E[X|R]] = \int_0^\infty \underbrace{E[X|R]}_{r} \cdot P_R(r)\,dr$ (15)

$= \int_0^\infty r \cdot P_R(r)\,dr = \frac{\alpha}{\lambda}$. (10)

Expected value of Gamma random variable